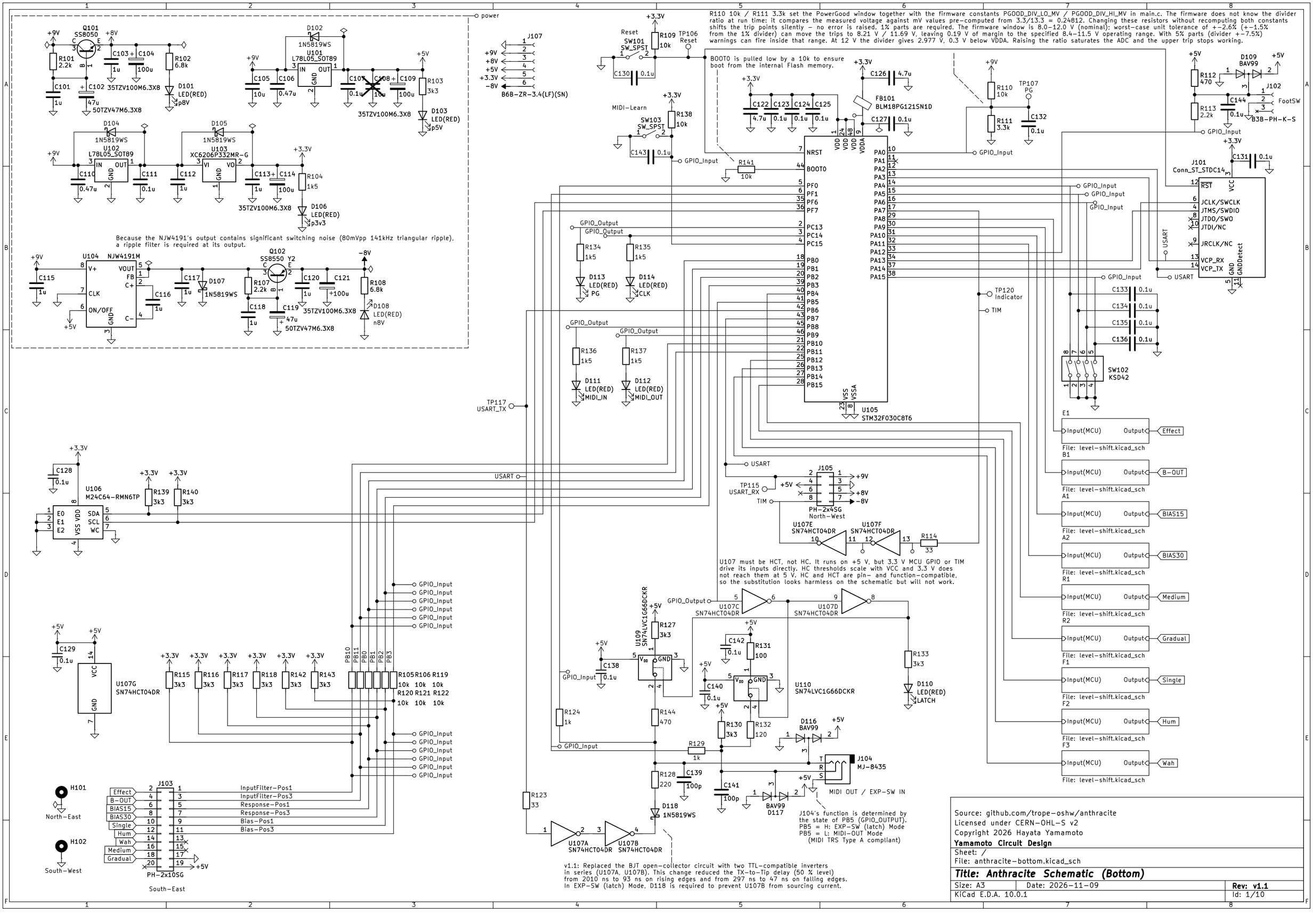
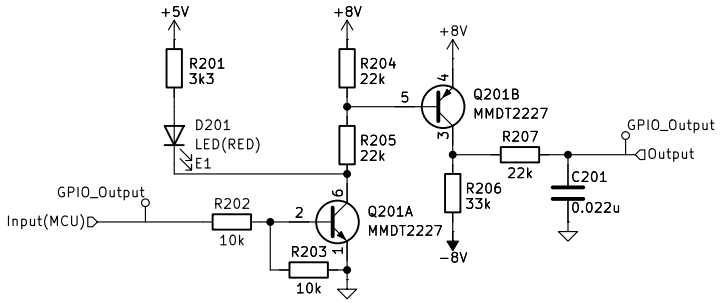

power
J107
B6B-ZR-3.4(LF)(SN)
Q101 SS8050
Q102 SS8550
U101 L78L05_SOT89
U102 L78L05_SOT89
U103 XC6206P332MR-G
U104 NJW4191M
D101 LED(RED) p8V
D103 LED(RED) p5V
D106 LED(RED) p3v3
D107 1N5819WS
D108 LED(RED) n8V
Because the NJW4191's output contains significant switching noise (80mVpp 141kHz triangular ripple), a ripple filter is required at its output.
R110 10k / R111 3.3k set the PowerGood window together with the firmware constants PGOOD_DIV_LO_MV / PGOOD_DIV_HI_MV in main.c. The firmware does not know the divider ratio at run time: it compares the measured voltage against mV values pre-computed from 3.3/13.3 = 0.24812. Changing these resistors without recomputing both constants shifts the trip points silently – no error is raised. 1% parts are required. The firmware window is 8.0–12.0 V (nominal): worst-case unit tolerance of +−2.6% (+−1.5% from the 1% divider) can move the trips to 8.21 V / 11.69 V, leaving 0.19 V of margin to the specified 8.4–11.5 V operating range. With 5% parts (divider +−7.5%) warnings can fire inside that range. At 12 V the divider gives 2.977 V, 0.3 V below VDDA. Raising the ratio saturates the ADC and the upper trip stops working.
BOOT0 is pulled low by a 10k to ensure boot from the internal Flash memory.
U105 STM32F030C8T6
J101 Conn_ST_STDC14
J102 FootSW B3B-PH-K-S
J103 PH-2x10SG
J104 MJ-8435
J105 PH-2x4SG North-West
SW101 SW_SPST
SW102 KSD42
SW103 SW_SPST
FB101 BLM18PG121SN1D
TP106 Reset
TP107 PG
TP120 Indicator
TP115 USART_RX
TP117 USART_TX
H101 North-East
H102 South-West
South-East
U106 M24C64-RMN6TP
U107 must be HCT, not HC. It runs on +5 V, but 3.3 V MCU GPIO or TIM drive its inputs directly. HC thresholds scale with VCC and 3.3 V does not reach them at 5 V. HC and HCT are pin- and function-compatible, so the substitution looks harmless on the schematic but will not work.
U107A SN74HCT04DR
U107B SN74HCT04DR
U107C SN74HCT04DR
U107D SN74HCT04DR
U107E SN74HCT04DR
U107F SN74HCT04DR
U107G SN74HCT04DR
U109 SN74LVC1G66DCKR
U110 SN74LVC1G66DCKR
D109 BAV99
D111 LED(RED) MIDI_IN
D112 LED(RED) MIDI_OUT
D113 LED(RED) PG
D114 LED(RED) CLK
D116 BAV99
D117 BAV99
D118 1N5819WS
D110 LED(RED) LATCH
MIDI OUT / EXP-SW IN
J104's function is determined by the state of PB5 (GPIO_OUTPUT).
PB5 = H: EXP-SW (latch) Mode
PB5 = L: MIDI-OUT Mode
(MIDI TRS Type A compliant)
v1.1: Replaced the BJT open-collector circuit with two TTL-compatible inverters in series (U107A, U107B). This change reduced the TX-to-Tip delay (50 % level) from 2010 ns to 93 ns on rising edges and from 297 ns to 47 ns on falling edges. In EXP-SW (latch) Mode, D118 is required to prevent U107B from sourcing current.
E1
B1
A1
A2
R1
R2
F1
F2
F3
File: level-shift.kicad_sch
Input(MCU)
Output
Effect
B-OUT
BIAS15
BIAS30
Medium
Gradual
Single
Hum
Wah
InputFilter-Pos1
InputFilter-Pos3
Response-Pos1
Response-Pos3
Bias-Pos1
Bias-Pos3
GPIO_Input
GPIO_Output
USART
TIM
Source: github.com/trope-oshw/anthracite
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Yamamoto Circuit Design
Sheet: /
File: anthracite-bottom.kicad_sch
Title: Anthracite Schematic (Bottom)
Size: A3
Date: 2026-11-09
Rev: v1.1
KiCad E.D.A. 10.0.1
Id: 1/10



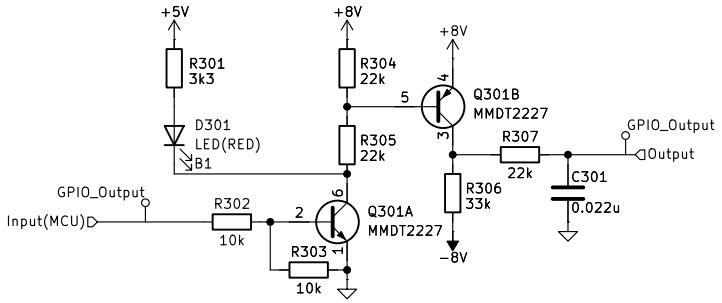
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Sheet: /E1/
File: level-shift.kicad_sch

Title: Anthracite Schematic (Bottom)

Size: A4	Date: 2026-11-09	**Rev: v1.1**
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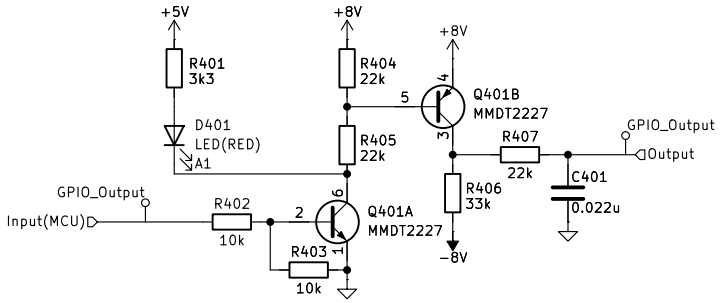
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Sheet: /B1/
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Title: Anthracite Schematic (Bottom)

Size: A4	Date: 2026-11-09	**Rev: v1.1**
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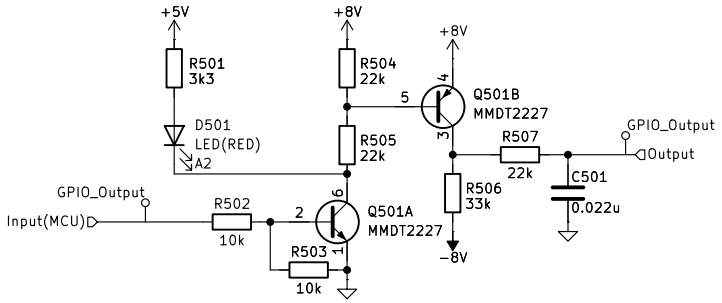
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Sheet: /A1/
File: level-shift.kicad_sch

Title: Anthracite Schematic (Bottom)

Size: A4	Date: 2026-11-09	**Rev: v1.1**
KiCad E.D.A. 10.0.1		Id: 4/10



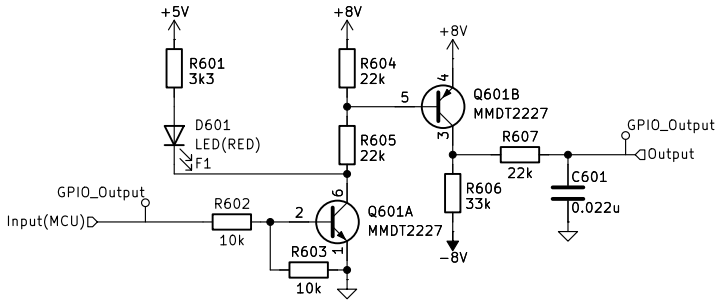
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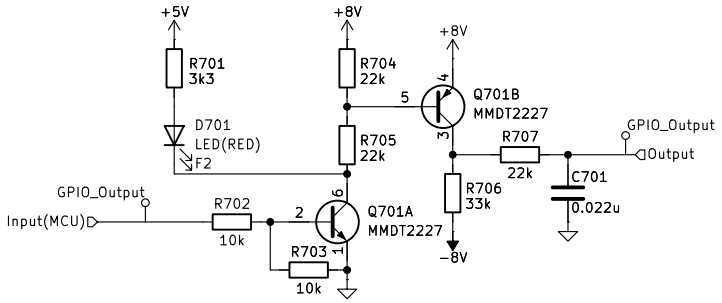
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File: level-shift.kicad_sch

Title: Anthracite Schematic (Bottom)

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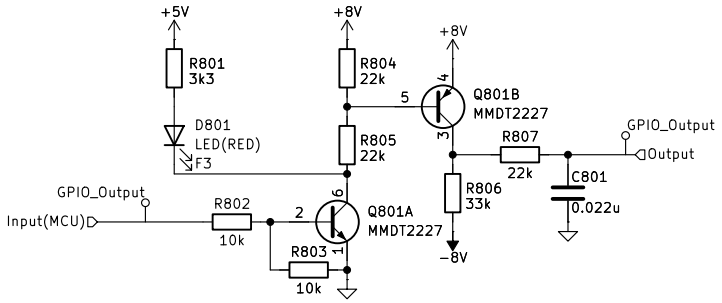
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File: level-shift.kicad_sch

Title: Anthracite Schematic (Bottom)

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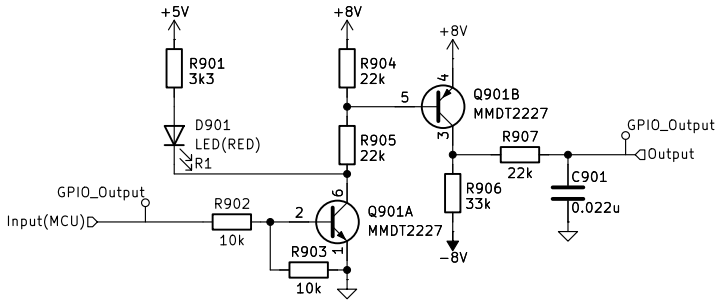
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File: level-shift.kicad_sch

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Size: A4	Date: 2026-11-09	**Rev: v1.1**
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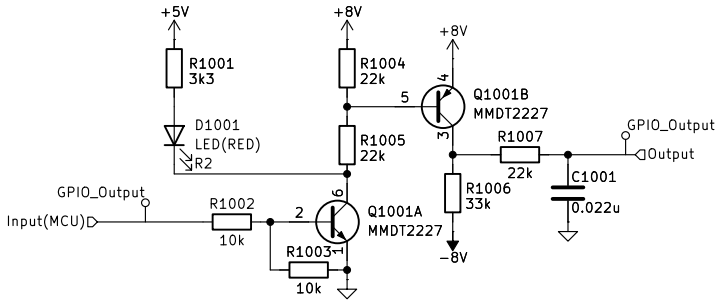
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